Husnu S. Narman

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ACADEMIC EXPERIENCE

Associate Professor: Marshall University — **Aug 2023 - Present**
Department of Computer Science — *Huntington WV, USA*

Assistant Professor: Marshall University — **Aug 2017 - May 2023**
Department of Computer Science — *Huntington WV, USA*

Instructor: Marshall University — **Jan 2017 - May 2017**
Department of Computer Science — *Huntington WV, USA*

Postdoctoral Fellow: Clemson University — **May 2016 - Dec 2016**
Department of Electrical and Computer Engineering — *Clemson SC, USA*

EDUCATION

Ph.D. in Computer Science — **May 2016**
University of Oklahoma — *Norman OK, USA*

M.S. in Computer Science — **May 2011**
University of Texas at San Antonio — *San Antonio TX, USA*

B.S. in Mathematics — **Aug 2006**
Abant Izzet Baysal University — *Bolu, Turkey*

HONORS

John and Frances Rucker Outstanding Graduate Advisor Award
Marshall University — *2025 - 2026*

Weisberg Faculty Research Award for Senior Faculty
College of Engineering and Computer Sciences, Marshall University — *2024 - 2025*

Mentorship Excellence Award
NSF S-STEM Project Work Studio, Marshall University — *2024 - 2025*

Weisberg Service Award
College of Engineering and Computer Sciences, Marshall University — *2021 - 2022*

Distinguished Artists and Scholars Junior Category Award
Marshall University — *2020 - 2021*

Weisberg Academy of Distinguished Teachers Award
College of Engineering and Computer Sciences, Marshall University — *2020 - 2021*

Outstanding Service Award
IEEE/ACM IoTDI — *2021*

Nomination for Pickens Queen Teaching Award
Marshall University — *2018, 2021*

Outstanding Ph.D. Student in Computer Science
University of Oklahoma — *2015 - 2016*

Study Abroad Fellowship for Higher Education
Turkish Ministry of National Education — *2007 - 2016*

Graduation as High Honor Student
Abant Izzet Baysal University — *2006*

RESEARCH INTEREST

Image Recognition: *Identify and classify objects and scenes in images and videos to solve various problems such as drone safety and structure inspections.*

Advanced Learning Technologies: *Investigate usage of high-tech products such as Virtual and Augmented Reality based applications with their effects in learning for K-12 and higher education.*

Data Mining: *Using sentiment analysis tools to understand the user behaviors for various applications like shopping habits and chain effects.*

Resource Allocation: *Determine scheduling models and allocations policies in Cloud and Fog Computing, IoT, and Networks with their applicable cases such as Vehicular Networks and Crowdsourcing.*

Smart Health: *Develop and investigate IoT products and mobile applications with machine learning algorithms to make the health system more accessible and productive by helping doctors and patients.*

R&D GROUP ATTAINMENTS

Fellowship & Scholarship

- Joshua Streets (Undergraduate): Marshall University Research and Creative Discovery Award for Summer 2026; *Project: Marshall Utility Data Analysis System*
- Josh Hatfield (Graduate): Marshall University Research and Creative Discovery Award for Spring 2026; *Project: Optimizing Skill Acquisition in Immersive Industrial Training Environments through Hierarchical Reinforcement Learning and Procedural Content Generation*
- Van Trung Le (Undergraduate): Marshall University Research and Creative Discovery Award for Spring 2026; *Project: A Low-Cost Vision-Based System for Railroad Gauge Measurement Using the Intel RealSense D435i Camera*
- Brandon Redden (Undergraduate): Marshall University Research and Creative Discovery Award for Summer 2025; *Project: Early Detection of Forest Fires Using Machine Learning*
- Connor Stonestreet (Undergraduate): Marshall University Research and Creative Discovery Award for Spring 2025; *Project: Automated Detection of Track Gauge Deviations Using Video and Depth Cameras with Machine Learning*
- Cade Parlato (Undergraduate): Marshall University SURE Summer Research Fellowship 2024; *Project: Calculating Aboveground Forest Biomass using Machine Learning with Image Segmentation*
- Andrew D'Arms (Undergraduate): Marshall University Research and Creative Discovery Award for Summer 2024; *Project: Timely Measuring of Earthquake Effects on Infrastructures According to Drone Sensor Fusion*
- Cade Parlato (Undergraduate): Marshall University Research and Creative Discovery Award for Spring 2024; *Project: Determining Tree Biomass in Forests using YOLOv7 Instance Segmentation*
- Josh Maddy (Undergraduate): NASA Undergraduate Research Fellowship for Spring 2023; *Project: The Metaphysical Exhibition An Exploration of Technology, the Arts, and Sciences: Expansion*
- Neil Loftus (Undergraduate): NASA Undergraduate Research Fellowship for Spring 2023; *Project: Detecting Birds with Real Time Image Processing for Drone Safety: Expansion*
- Neil Loftus (Undergraduate): NASA Undergraduate Research Fellowship 2022 - 2023; *Project: Detecting Birds with Real Time Image Processing for Drone Safety*
- Neil Loftus (Undergraduate): Marshall University Research and Creative Discovery Award for Summer 2022; *Project: The Cybersecurity Packet Control Simulator: The Effect of Visual Learning Tools on Retention of Information in Computer Science*
- Josh Maddy (Undergraduate): Marshall University Research and Creative Discovery Award for Summer 2022; *Project: The Metaphysical Exhibition an Exploration of Technology, the Arts, and Sciences*
- Josh Maddy (Undergraduate): Marshall University SURE Summer Research Fellowship 2022; *Project: Augmented Reality as an Aid for Physics Concepts*
- Neil Loftus (Undergraduate): Marshall University SURE Summer Research Fellowship 2022; *Project: The Cybersecurity Packet Control Simulator*

- Eric Dillion (Undergraduate): Marshall University Research and Creative Discovery Award for Spring 2022; *Project: Automatic Feedback System to Teach Cybersecurity Principles*
- Eric Shoemaker (Undergraduate): Marshall University Research and Creative Discovery Award for Summer 2021; *Project: Crowdsourcing based Community Infrastructure Management Application*
- Eric Shoemaker (Undergraduate): NASA Undergraduate Research Fellowship for Spring 2021; *Project: Community Infrastructure Management Application*
- Jarred Carter (Undergraduate): NASA Undergraduate Research Fellowship 2020 - 2021; *Project: Simulation for Trade-off Model of Fog-Cloud Computing for Space Network*
- William Coleman (Undergraduate): Marshall University SURE Summer Research Fellowship 2020; *Project: Enhancing STEM Education with Augmented Reality*
- James Farley (Undergraduate): NASA Undergraduate Research Fellowship for Spring 2020; *Project: Emotion Classification of Users in Social Media*
- Alex Canfield, Cameron Berry, Jeremy Giese, Logan Carpenter (Undergraduates): Marshall University Research and Creative Discovery Award for Spring 2019; *Project: Data Structure with Augmented Reality*
- James Farley (Undergraduate): Marshall University Research and Creative Discovery Award for Fall 2019; *Project: Emotion Classification of Users based on the Comments and Emojis in Social Media*
- Jarred Carter (Undergraduate): NASA Undergraduate Research Fellowship 2019 - 2020; *Project: Trade-off Model of Fog-Cloud Computing for Space Network*
- Caleb Kesler (Undergraduate): Marshall University Research and Creative Discovery Award for Summer 2019; *Project: Development of Story-Assisted Platform for Early Childhood for Coding*
- Alymbek Damir Uulu (Undergraduate): Marshall University SURE Summer Research Fellowship 2019; *Project: Profile Analysis on Cryptocurrency Investors and Social Engineering on Their Prices*
- Jared Lee Lewis (Undergraduate): Marshall University Research and Creative Discovery Award for Fall 2018; *Project: Automated IP Reputation Analyzer System with Machine Learning*
- Geanina Tambaliuc (Undergraduate): Marshall University SURE Summer Research Fellowship 2018; *Project: Automated IP Reputation Analyzer System*
- Alex Kacinari, Chris Murphy, Derek M Staley (Undergraduates): Marshall University Research Scholar Award for Spring 2018; *Project: Suturing Technique Simulation*
- Charlie Murphy, Michael B Branard, Steven D. Gunnels (Undergraduates): Marshall University Research Scholar Award for Spring 2018; *Project: Embedded Storybook Game*

Thesis and Capstone Advising

- Hwapyeong Song: PhD Advisor
- Shahid Ali: PhD Committee Member
- Josh Hatfield: Master Thesis Co-Advisor
- Omar Al Babele: Master Thesis Committee Chair
- George Urling: Master Thesis Committee Chair
- Neil Loftus: Master Thesis Committee Chair
- Anastasiia Sukhanova: Master Thesis Committee Member
- Raghad Alhusari: Master Thesis Committee Member
- Eric Shoemaker: Master Thesis Committee Member
- Vishwanshi Joshi: Master Thesis Committee Chair
- Yucheng Li: Master Thesis Committee Member
- Govind Yatnalkar: Master Thesis Advisor
- Craig Carpenter II, David Wills, Eric Dillon, and John Cook (Capstone Advisor)
- Geanina Tambaliuc, Hwapyeong Song, and Wesly Webb (Capstone Advisor)
- Alex Canfield, Cameron Berry, Jeremy Giese, and Logan Carpenter (Capstone Advisor)
- Alex Kacinari, Chris Murphy, and Derek M Staley (Capstone Advisor)
- Charlie Murphy, Michael B. Branard, and Steven D. Gunnels (Capstone Advisor)

Internship Advisor (FMIP)

- Hwapyeong Song (Graduate): WV Department of Education Internship and Paving Academy 2020

- Neil Loftus (High School): Paving Academy 2020
- Geanina Tambaliuc (Undergraduate): WV Department of Education Internship 2019
- Anh Nguyen (Undergraduate): WV Department of Education Internship 2019
- Jake Gressang (Undergraduate): WV Department of Education Internship 2019
- Kuo Chi Fang (Graduate): WV Department of Education Internship 2018
- Ibrahim Hussein Mwinyi (Graduate): WV Department of Education Internship 2018

Publications

- High School Students: Isabella Schrader, Laina Karim, Neil Loftus, Sawyer Slack.
- Undergraduate Students: Advay Chandramouli, Alex Canfield, Alymbek Damir Uulu, Amelia McGinty, Andrew D'Arms, Anh Nguyen, Aayush Damai, Cade Parlato Cameron Berry, Cameron Green, Connor Stonestreet, Craig Carpenter II, Eric M. Dillon, Eric Shoemaker, Furkan Kizilay, Geanina Tambaliuc, Greg Weed, Hannah Vitalos, James Farley, Jared Lee Lewis, Jarred Carter, Jeremy Giese, John Cook, Josh Maddy, Logan Carpenter, Mina R. Narman, Mingyan Liu, Neil Loftus, Thomas D. Wills, Van Trung Le, William Coleman.
- Graduate Students: Abdullah Jawad, Alexander Lambert, Eric Shoemaker, Govind Yatnalkar, Hwapyeong Song, Josh Hatfield, Ibrahim Hussein Mwinyi, Kanimozhi Kalaichelavan, Kuo Chi Fang, Noah Quesenberry, Sreehari Sreenath.

TAUGHT COURSES AND EFFECTIVENESS

- Data Structures and Algorithms (Spring'17 - 26, Fall'17 - 26)
- Advanced Database Systems (Spring'17 - 18, 21 - 26)
- Cloud Computing (Fall'17 - 18, 20 - 22, 26)
- Computer Science II - OOP Java (Spring'19 - 20, '24 - 25, Fall'19, '23 - 25)
- Advanced Data Structures and Algorithms (Spring'17)
- Automata and Formal Languages (Fall'18)
- Database Engineering (Spring'17)
- Database Management (Spring'17 - 18)
- Cyberwarfare (Spring'20 - 22)
- Geometry (Fall'06, Spring'07)

Q1	I believe that I learned in this class.
Q2	The course was well organized.
Q3	This course challenged me intellectually.
Q4	I have become more competent in this area because of this course.
Q5	The objectives of the course were well explained.
Q6	The instructor followed his/her syllabus.
Q7	The instructor gave clear explanations to clarify concepts.
Q8	The instructor was supportive in academic situations.
Q9	The instructor showed enthusiasm when teaching.
Q10	The instructor informed students of their progress.
Q11	The instructor's use of examples helped to get points across in class.
Q12	The instructor adequately explained the grading scale.
Q13	The instructor treated me fairly.
Q14	The instructor was enthusiastic about the course material.
Q15	The instructor encouraged students to ask questions.
Q16	The instructor provided me with an effective array of challenges.
Q17	The instructor carefully answered questions raised by students.
Q18	The instructor treated students with respect.
Q19	The instructor presented material in a clear manner.
Q20	The instructor used class time well.
Q21	The instructor seemed genuinely interested in wanting me to learn.
Q22	I would recommend this instructor to other students.

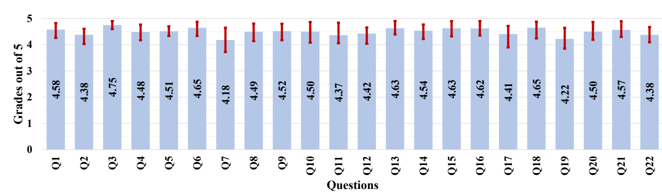

Figure 1: Each question in the left table was rated by over 720 undergraduate and graduate students on a scale from 1 to 5 during course evaluations. The figure's bars display the average scores across all semesters, along with the minimum and maximum semester-based averages for each question. (All ratings are based on a 5-point scale.)

SERVICES

Marshall University **Huntington WV, USA**

- Organization of Computer Science Adventure Zone Summer Camp 2017 - Present
- Elected College Representative on the Athletic Committee 2019 - Present

- Faculty Search Committee Member and Chair 2023 - Present
- CS Freshman Orientation 2018 - Present
- CS Online Course Development 2018 - Present
- CS Conference Organization Committee 2018 - Present
- Computer Science (CS) Student Leadership Program Coordinator 2019 - 2020
- CS Representative on the College Outreach and Recruitment Committee 2019 - 2020
- CS Outreach Committee Chair 2017 - 2020
- CS Faculty Candidate Interviewing Committee 2017 - 2019
- Catalog Editing and Correction for BS and MS Programs in CS 2017 - 2018
- CS Project Room Management 2017 - 2018

PROFESSIONAL ACTIVITIES

Co-Lead for Training Programs **2021 - Present**
Institute for Cybersecurity (ICS), Marshall University

Editorial Board **2018 - Present**
Editorial Board: Elsevier Journal of Network and Computer Applications

Summer Camp on Robotics and Cybersecurity for K-12 **2017 - Present**
Marshall University *Huntington WV, USA*

Reviewer **2023 - Present**
National Science Foundation (NSF)

Faculty Mentor Internship Program **2018 - 2020**
WV Department of Education *Huntington WV, USA*

Virtual Conference Working Group Lead
• 2021 ACM/IEEE Conference on Internet of Things Design and Implementation

Publicity Co-Chair
• 2017 International Conference on Networking, Architecture, and Storage • 2017 International Conference on Computer Communications and Networks

Technical Program Committee
• IEEE Global Communications Conf. • IEEE Int. Conf. on Communications • Springer Ubiquitous Networking Conf. • IEEE Wireless Communications and Networking Conf. • IEEE Int. Conf. on Communications, Network, and Satellite • IEEE 5G World Forum (WF-5G) • IEEE Int. Conf. on Internet of Things and Intelligence System • Elsevier Int. Conf. on Ambient Systems, Networks and Technologies • IEEE Int. Conf. on Fog and Edge Computing • IEEE Symp. on Signal Processing and Information Technology • IEEE Int, Conf. on Signals and Systems • IEEE/ACM Int. Symp. in Cluster, Cloud, and Grid Computing • IEEE Middle East & North Africa Communications Conf. • IEEE Int. Conf. on Wireless Networks and Mobile Communications • IEEE TENCON

Journal Reviewer
• IEEE Journal on Selected Areas in Communications • IEEE Transactions on Mobile Computing • IEEE Transactions on Parallel and Distributed Systems • IEEE Transactions on Vehicular Technology • IEEE Transactions on Industrial Electronics • IEEE Transaction on Intelligent Transportation Systems • IEEE Transactions on Sustainable Computing, • ACM Transactions on Cyber-Physical Systems • ACM Transactions on Knowledge Discovery from Data • Elsevier Journal of Network and Computer Applications • Elsevier Future Generation Computer System • MDPI Sensors • Wiley Software: Practice and Experience

Membership
• IEEE Senior Member (2022 - Present) • IEEE Member (2014 - 2021)

VOLUNTEER ACTIVITIES

VEX IQ Robot Tournament for Middle and Elementary Schools **2019 - Present**
Marshall University *Huntington WV, USA*

Faculty adviser for IEEE Marshall University Student Chapter **2022 - Present**

Marshall University — *Huntington WV, USA*

Faculty adviser for IEEE Women in Engineering — **2024 - Present**
Marshall University — *Huntington WV, USA*

Faculty adviser for Vex Robotics Club — **2024 - Present**
Marshall University — *Huntington WV, USA*

Faculty adviser for Marshall University Technology Association — **2018 - Present**
Marshall University — *Huntington WV, USA*

Coordinator and Initiator for Community Service Leadership Program — **2019 - 2020**
Marshall University — *Huntington WV, USA*

CERTIFICATION

Responsible Conduct of Research — **2018**
Collaborative Institutional Training Initiative Program — *Huntington WV, USA*

Behavioral & Social Science Research — **2018**
Collaborative Institutional Training Initiative Program — *Huntington WV, USA*

Independent Improving Your Online Course (IYOC) — **2017**
Quality Matters — *Huntington WV, USA*

Java SE 7 — **2016**
Robert Half Technology — *Clemson SC, USA*

GRANTS

External Funded Grants: Total Amount:$4,665,550

[18] *REU Site: Undergraduate Research in Data Analytics (URDA)*. **Senior Personnel**. NSF, 2025. Amount: $388K, **Funded**.

[17] *Automatic Assessment and Repair for Railroads*. **Co-PI**. ERDC, 2024. Amount: $2M, **Funded**.

[16] *Establishing a Cyber Security Center for Critical Infrastructure*. **Responsible for Training Modules**. Department of Education, 2023. Amount: Total:$1,5M, Share: $750K, **Funded**.

[15] *Detecting Birds with Real Time Image Processing for Drone Safety*. **Mentor**. NASA West Virginia Undergraduate Research Fellowship, 2022. Amount: $5K, **Funded**.

[14] *Detecting Birds with Real Time Image Processing for Drone Safety: Expansion*. **Mentor**. NASA Undergraduate Affiliate Fellowship Program, 2022. Amount: $1K, **Funded**.

[13] *Experiences for Teachers on Cyber Security*. **Proposal Developer**. NSF/NSA GenCyber Teacher Summer Camp, 2022. Amount: $140K, **Funded**.

[12] *The Metaphysical Exhibition an Exploration of Technology, the Arts, and Sciences*. **Mentor**. NASA Undergraduate Affiliate Fellowship Program, 2022. Amount: $1K, **Funded**.

[11] *VEX IQ Workshops for Students and Staff of Playmates*. **Responsible for Summer Workshops**. Department of Education, 2022. Amount: Total:$1,5M, Share: $40K, **Funded**.

[10] *Crowdsourcing Infrastructure Management System*. **Mentor**. NASA Undergraduate Affiliate Fellowship Program, 2021. Amount: $1K, **Funded**.

[9] *Community Infrastructure Management Application*. **Mentor**. NASA Undergraduate Affiliate Fellowship Program, 2020. Amount: $1K, **Funded**.

[8] *Development of the Pavement Preservation and Rehabilitation Academy*. **Co-PI**. Wirtgen Group - John Deere Company, 2020. Amount: $40K, **Funded**.

[7] *Scholarships and a Project-based Work Studio to Support Undergraduate Student Graduation and Entry into Computer Science, Engineering, and Safety Technology Careers*. **Co-PI**. NSF, 2020. Amount: $1M, **Funded**.

[6] *Trade-off Model of Fog-Cloud Computing for Space Network*. **Mentor**. NASA West Virginia Undergraduate Research Fellowship, 2020. Amount: $5K, **Funded**.

[5] *Adventure Zone Teacher Academy on Cybersecurity*. **PI**. NSF/NSA GenCyber Teacher Summer Camp, 2019. Amount: $73K, **Funded**.

[4] *Emotion Classification of Users in Social Media*. **Mentor**. NASA Undergraduate Affiliate Fellowship Program, 2019. Amount: $1K, **Funded**.

[3] *Trade-off Model of Fog-Cloud Computing for Space Network*. **Mentor**. NASA West Virginia Undergraduate Research Fellowship, 2019. Amount: $5K, **Funded**.

[2] *WV Faculty Mentor Internship Program - Calendar and Special Education Applications*. **Co-PI**. West Virginia Department of Education, 2019. Amount: $85K, **Funded**.

[1] *WV Faculty Mentor Internship Program - Complain Management Application and Security of Applications*. **Co-PI**. West Virginia Department of Education, 2019. Amount: $79K, **Funded**.

Internal Funded Grants: Total Amount: $197,150

[36] *A Low-Cost Vision-Based System for Railroad Gauge Measurement Using the Intel RealSense D435i Camera*. **Mentor**. Undergraduate Spring Research and Creative Discovery, Marshall University, 2026. Amount: $2.5K, **Funded**.

[35] *Marshall Utility Data Analysis System*. **Mentor**. Undergraduate Summer Research and Creative Discovery, Marshall University, 2026. Amount: $5K, **Funded**.

[34] *Optimizing Skill Acquisition in Immersive Industrial Training Environments through Hierarchical Reinforcement Learning and Procedural Content Generation*. **Mentor**. Graduate Spring Research and Creative Discovery, Marshall University, 2026. Amount: $2.5K, **Funded**.

[33] *Early Detection of Forest Fires Using Machine Learning*. **Mentor**. Undergraduate Summer Research and Creative Discovery, Marshall University, 2025. Amount: $5K, **Funded**.

[32] *Quinlan Endowment Faculty Travel*. **PI**. Quinlan Endowment Faculty Travel, Marshall University, 2025. Amount: $300.00, **Funded**.

[31] *Automated Detection of Track Gauge Deviations Using Video and Depth Cameras with Machine Learning*. **Mentor**. Undergraduate Spring Research and Creative Discovery, Marshall University, 2024. Amount: $1,750.00, **Funded**.

[30] *Calculating Aboveground Forest Biomass using Machine Learning with Image Segmentation*. **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2024. Amount: $4K, **Funded**.

[29] *Enhancing Programming Education through Artificial Intelligence-Driven Tools*. **PI**. Faculty Summer Research, Marshall University, 2024. Amount: $2K, **Funded**.

[28] *Quinlan Endowment Faculty Travel*. **PI**. Quinlan Endowment Faculty Travel, Marshall University, 2024. Amount: $500.00, **Funded**.

[27] *Timely Measuring of Earthquake Effects on Infrastructures According to Drone Sensor Fusion*. **Mentor**. Undergraduate Summer Research and Creative Discovery, Marshall University, 2024. Amount: $5K, **Funded**.

[26] *Determining Tree Biomass in Forests using YOLOv7 Instance Segmentation*. **Mentor**. Undergraduate Spring Research and Creative Discovery, Marshall University, 2023. Amount: $1,750.00, **Funded**.

[25] *Augmented Reality as an Aid for Physics Concepts*. **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2022. Amount: $4K, **Funded**.

[24] *Auto Feedback System based on Artificial Intelligence for Cybersecurity Learners*. **PI**. Faculty Summer Research, Marshall University, 2022. Amount: $2K, **Funded**.

[23] *Automatic Feedback System to Teach Cybersecurity Principles*. **Mentor**. Undergraduate Fall Research and Creative Discovery, Marshall University, 2022. Amount: $1,750.00, **Funded**.

[22] *Marshall University Presentation Center: Developing A Campus-Wide Resource to Enhance Communication Fluency Across the Curriculum*. **Co-PI**. Hedrick Program Grant for Teaching Innovation, Marshall University, 2022. Amount: $5K, **Funded**.

[21] *The Cybersecurity Packet Control Simulator*. **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2022. Amount: $4K, **Funded**.

[20] *The Cybersecurity Packet Control Simulator: The Effect of Visual Learning Tools on Retention of Information in Computer Science*. **Mentor**. Undergraduate Summer Research and Creative Discovery, Marshall University, 2022. Amount: $5K, **Funded**.

[19] *The Metaphysical Exhibition an Exploration of Technology, the Arts, and Sciences*. **Mentor**. Undergraduate Summer Research and Creative Discovery, Marshall University, 2022. Amount: $5K, **Funded**.

[18] *Crowdsourcing Infrastructure Management System*. **Mentor**. Undergraduate Summer Research and Creative Discovery, Marshall University, 2021. Amount: $5K, **Funded**.

[17] *Enhancing STEM Education with Augmented Reality*. **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2020. Amount: $4K, **Funded**.

[16] *A Smart Therapy Tool for Feeding and Speech Disorder Detection*. **PI**. John Marshall University Summer Scholars Awards, Marshall University, 2019. Amount: $6.5K, **Funded**.

[15] *Artificial Intelligence and Integrated Fog-Cloud Computing based Matching Algorithm for Carpooling*. **PI**. Faculty Summer Research, Marshall University, 2019. Amount: $2K, **Funded**.

[14] *Augmented Reality based Application for Data Structure*. **Mentor**. Undergraduate Fall Research Scholar Awards, Marshall University, 2019. Amount: $250.00, **Funded**.

[13] *Development of Story-Assisted Platform for Early Childhood for Coding*. **Mentor**. Undergraduate Summer Research and Creative Discovery, Marshall University, 2019. Amount: $5K, **Funded**.

[12] *Emotion Classification of Users based on the Comments and Emojis in Social Media*. **Mentor**. Undergraduate Fall Research and Creative Discovery, Marshall University, 2019. Amount: $1,750.00, **Funded**.

[11] *Profile Analysis on Cryptocurrency Investors and Social Engineering on their Prices*. **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2019. Amount: $4K, **Funded**.

[10] *Quinlan Endowment Faculty Travel*. **PI**. Quinlan Endowment Faculty Travel, Marshall University, 2019. Amount: $500.00, **Funded**.

[9] *Student Research and Innovation Center*. **Co-PI**. Cross-disciplinary Research Facilitation Grant, Marshall University, 2019. Amount: $500.00, **Funded**.

[8] *Automated IP Reputation Analyzer System*. **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2018. Amount: $4K, **Funded**.

[7] *Automated IP Reputation Analyzer System*. **Mentor**. Undergraduate Fall Research and Creative Discovery, Marshall University, 2018. Amount: $1,750.00, **Funded**.

[6] *Game Embedded Storybook*. **Mentor**. Undergraduate Spring Research Scholar Awards, Marshall University, 2018. Amount: $175.00, **Funded**.

[5] *Placement of Electric Vehicle Charging Sections and Traffic Management*. **PI**. Faculty Summer Research, Marshall University, 2018. Amount: $2K, **Funded**.

[4] *Quinlan Endowment Faculty Travel*. **PI**. Quinlan Endowment Faculty Travel, Marshall University, 2018. Amount: $400.00, **Funded**.

[3] *Suturing Technique Simulation*. **Mentor**. Undergraduate Spring Research Scholar Awards, Marshall University, 2018. Amount: $175.00, **Funded**.

[2] *Research Initiation and Developemnt*. **PI**. Marshall University, 2017. Amount: $100,000.00, **Funded**.

[1] *Self-dynamic Data Center Management to Optimize Search Speed for Media Type Files*. **PI**. Faculty Summer Research, Marshall University, 2017. Amount: $2,000.00, **Funded**.

PUBLICATIONS

$\pm$: High School Students, $*$: Undergraduate Students, $+$: Graduate Students.

Patents and Intellectual Property

[1] *Oral Therapy Tool, System, and Related Methods*. Patent Appl. Serial No. 62/947,264, Regular, United States. (Submitted: November 2019, Application: December 2019, Updated: October 2022, Allowed: January 2023, Granted: August 2023 – 11,712,366).

Books

[1] **Husnu S. Narman** and Mohammed Atiquzzaman. *Carrier Assignment and Packet Scheduling in LTE-A and Wi-Fi*. Dissertation as a Book. LAP LAMBERT Academic Publishing, May 2016, p. 160. ISBN: 9783659891977. URL: `https://www.amazon.com/Carrier-Assignment-Packet-Scheduling-Wi-Fi/dp/3659891975`.

Journals

[18] Shahid Ali[+], Ammar Alzarrad, Sudipta Chowdhury, **Husnu S. Narman**, and Shifa Saleem Ahmed Khan. "Benchmarking YOLOv8-YOLOv12 Models for Image-Space Near-Miss Detection in Construction Safety". In: *Frontiers in Built Environment* (2026).

[17] Shahid Ali[+], Ammar Alzarrad, Hwapyeong Song[+], and **Husnu S. Narman**. "BuildSafe: A Generative AI Framework for Joint Construction Safety, Permit, and Community-Impact Reasoning from Linked NYC DOB-OSHA-311 Data". In: *Frontiers in Built Environment* (2026).

[16] Van Trung Lee*, Hwapyeong Song[+], **Husnu S. Narman**, Pingping Zhu, Ammar Alzarrad, Abby Cisco, and Jeremy Beasley. "Automated Measurement of Horizontal Gauge Deviation in Railroads Using Depth Sensor Camera and Machine Learning". In: *IEEE Access* (2025).

[15] Furkan Kizilay, Mina R. Narman*, Hwapyeong Song[+], **Husnu S. Narman**, Cumhur Cosgun, and Ammar Alzarrad. "Evaluating Fine Tuned Deep Learning Models for Real-Time Earthquake Damage Assessment with Drone-Based Images". In: *AI in Civil Engineering* (2024).

[14] Neil Loftus* and **Husnu S. Narman**. "Use of Machine Learning in Interactive Cybersecurity and Network Education". In: *Sensors* 23.6 (2023). ISSN: 1424-8220.

[13] Amrit Pal, Abishi Chowdhury, Satakshi, **Husnu S. Narman**, Arkabandhu Chowdhury, and Manish Kumar. "Random Partition based Adaptive Distributed Kernelized SVM for Big Data". In: *IEEE Access* (2022).

[12] Arnob Paul, Md. Hasanul Islam, Md. Shohrab Hossain, and **Husnu S. Narman**. "A novel zone walking protection for secure DNS Server". In: *International Journal of Interdisciplinary Telecommunications and Networking* (2022).

[11] **Husnu S. Narman**, Haroon Malik, and Govind Yatnalkar[+]. "An Enhanced Ride Sharing Model Based on Human Characteristics, Machine Learning Recommender System, and User Threshold Time". In: *Springer Journal of Ambient Intelligence and Humanized Computing* (2021). **Invited**.

[10] Jinwei Liu, Haiying Shen, Hongmei Chi, **Husnu S. Narman**, Yongyi Yang, Long Cheng, and Wingyan Chung. "A Low-Cost Multi-Failure Resilient Replication Scheme with Data Correlation for High Data Availability in Cloud Storage". In: *IEEE/ACM Transaction on Networking* (2020).

[9] Abishi Chowdhury, Shital A. Raut, and **Husnu S. Narman**. "DA-DRLS: Drift adaptive deep reinforcement learning based scheduling for IoT resource management". In: *Journal of Network and Computer Applications* 138 (May 2019), pp. 51–65.

[8] Ankur Sarker, Haiying Shen, M. Rahman, M. Chowdhury, K. Dey, F. Li, Y. Wang, and **Husnu S. Narman**. "A Review of Sensing and Communication, Human Factors, and Controller Aspects for Information-Aware Connected and Automated Vehicles". In: *IEEE Transactions on Intelligent Transportation Systems* (March 2019).

[7] Kuo-Chi Fang[+], **Husnu S. Narman**, Ibrahim Hussein Mwinyi[+], and Wook-Sung Yoo. "PPHA-Popularity Prediction Based High Data Availability for Multimedia Data Center". In: *International Journal of Interdisciplinary Telecommunications and Networking* 11.1 (January 2019), pp. 17–29.

[6] Jinwei Liu, Haiying Shen, **Husnu S. Narman**, Z. Lin, and Z. Li. "Popularity-aware Multi-failure Resilient and Cost-effective Replication for High Data Durability in Cloud Storage". In: *IEEE Transactions on Parallel and Distributed Systems* (October 2018).

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